

Extraction of zinc

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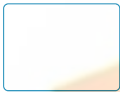
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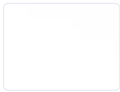


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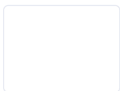
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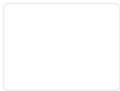


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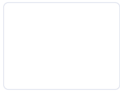
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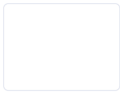
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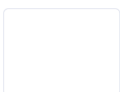
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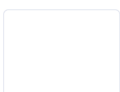
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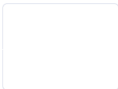
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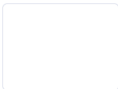
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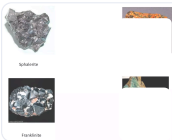
Extraction of Zn

Presented by
Nik
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LKS

07/27

Introduction

- Zinc is mainly used for
Galvanizing
Production of alloys
Dry cell batteries
- Its minerals are
Sphalerite (ZnS)
Zincite (ZnO)
Smithsonite ($ZnCO_3$)
Franklinite ($ZnO \cdot FeMn_2O_4$)



AD

Pyro metallurgical extraction of Zinc



Procedures

- The ore is first ground and subjected to float
- The concentrate obtained is subjected to roast
- Roasted product is ground and agglomerated to a 10mm size for smelting and sintering

$$\text{ZnS} + 1.5\text{O}_2 \rightarrow \text{ZnO} + \text{SO}_2 \quad (\text{re})$$
$$\text{ZnO} + \text{CO} \rightarrow \text{Zn} (\text{g}) + \text{CO}_2 \quad (\text{re})$$
$$\text{C} + \text{CO}_2 \rightarrow 2\text{CO} \quad (\text{re})$$

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- ZnO is reduced by carbon
- Make sure CO partial pressure is maintained
- Zinc vapor obtained is condensed at 450°C
- Zinc condensation has to be carried out quickly so previous formation of base oxide of Zn (ZnO)

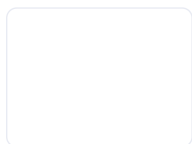
Hydro metallurgical extraction of zinc



The flowchart illustrates the hydro metallurgical extraction of zinc. It begins with 'Zinc concentrate' (ZnS), which is 'Roasted' to produce 'Zinc oxide' (ZnO). This ZnO is then 'Reduced by carbon' to form 'Zinc metal' (Zn). The Zn metal is 'Leached' with 'Sulfuric acid' (H2SO4) to produce 'Zinc sulfate' (ZnSO4). This is followed by 'Evaporation' to yield 'Zinc sulfate crystals' (ZnSO4 · 7H2O). Finally, 'Zinc sulfate crystals' are 'Electrolyzed' to produce 'Zinc metal' (Zn).

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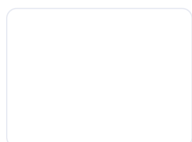
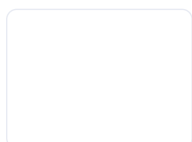
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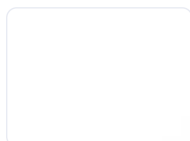
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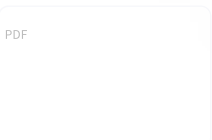
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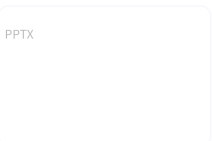
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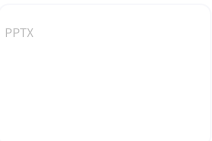
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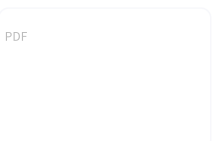
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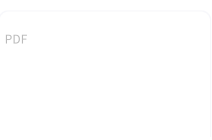
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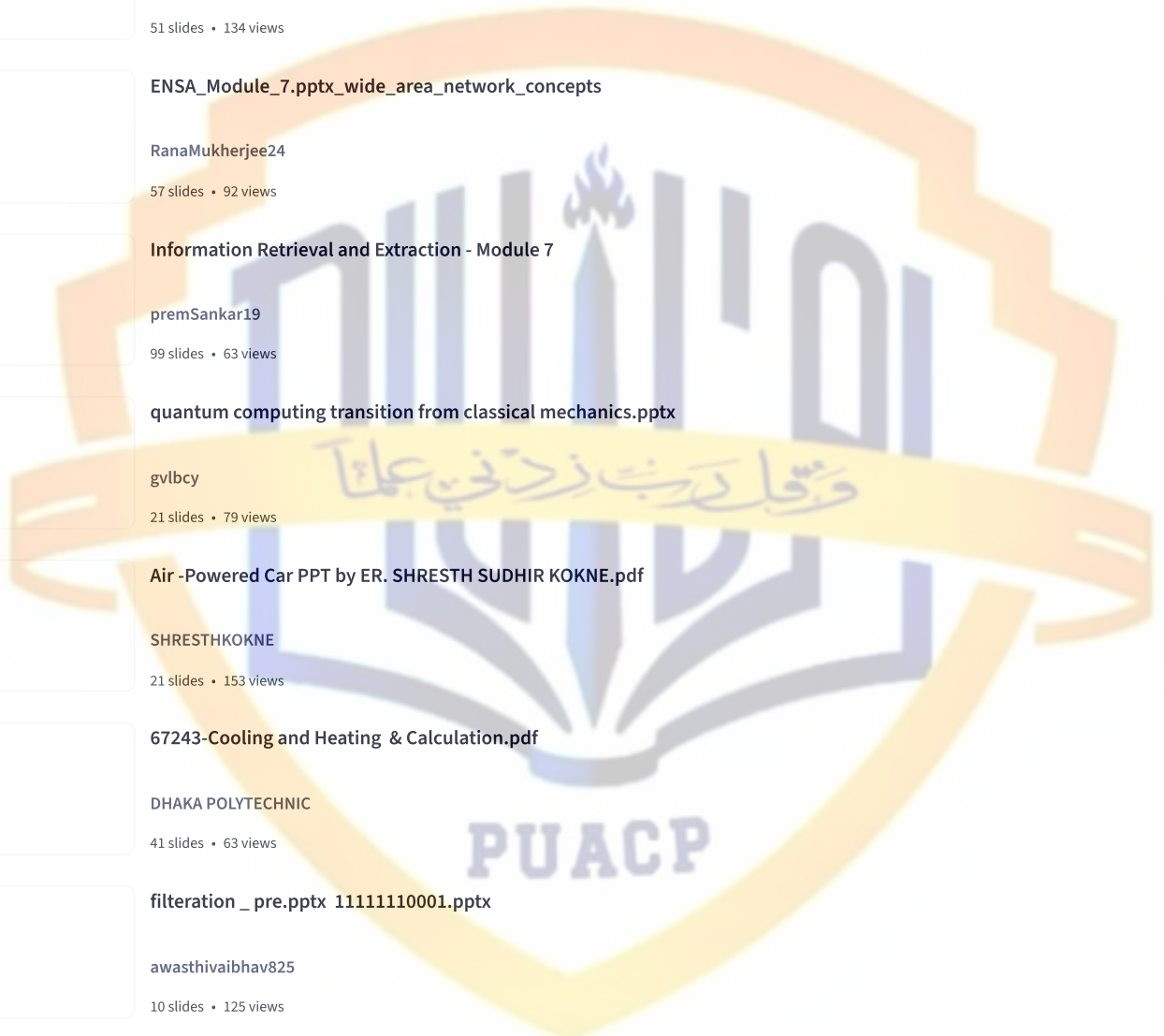
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Extraction of zinc

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2. **Introduction • Zinc** is mainly used for Galvanizing Production of alloys Dry cell batteries • Its minerals are Sphalerite (ZnS) Zincite (ZnO) Smithsonite (ZnCO₃) Franklinite (ZnO(Fe,Mn)₂O₃)
3. **Sphalerite Zincite Franklinite** Smithsonite
4. **Pyro metallurgical extraction** of Zinc Jkkkkkkkkkkk
5. **Procedures • The** ore is first ground and subjected to froth flotation • The concentrate obtained is subjected to roasting • Roasted oxide is ground and agglomerated and then sintered at 1200 in a Dwight Lloyd sintering and sent for retort reduction
6. **• ZnO is** reduced by carbon • Make sure CO partial pressure is maintained to prevent reverse reaction • Zinc vapour obtained is condensed at 4500c to form liquid Zn • Zinc condensation has to be carried out quickly to prevent formation of blue oxide of Zn+ZnO
7. **Hydro metallurgical extraction** of zinc
8. **Procedures • Dissolution** of zinc calcine in sulphuric acid • Precipitation of Cd by adding zinc dust to the leach solution i.e cementation • Precipitation of cobalt from leach solution by adding alpha nitroso – beta naphthol • Adjust the pH of leach solution to 5.3 to hydrolyze Fe and Al and add of MnO₂ to oxidize iron to trivalent state to produce a precipitate that can absorb As, Sb, and Ge • Do electrolysis of filterate solution
9. **Imperial Smelting Process** J,h
10. **Advantages of ISP** • Its possible to simultaneously smelt low grade complex mixed charges to recover both Zn and Pb • Recovery of Zn is less expensive • A wide variety of furnace sizes are available • Furnace operation is fully automated • The mechanism is highly robust
11. **Reactions Involved**
12. **Procedures • Preheated** coke and sinter is charged from the top of the furnace • The lead becomes liquid metal and flows to the bottom of the furnace collecting precious metals like gold , silver etc (lead bullion) • Slag and lead bullion at the bottom can be separated because of their density differences • The furnace gases produced flows upwards to the condensers • Zinc and lead are collected after condensation
14. **Procedures • The** condensers perform two functions ie cool the furnace gases, help in dissolving the condensed Zn in molten lead • The rotors produce a spray of lead bath • Zn is cooled rapidly and goes in to the solution with lead • Zinc containing lead is pumped out of from the condensers and cooled • During cooling the lead becomes saturated with zinc and excess zinc emerges from the lead and floats on the surface • Zinc is separated and the lead is fed back to the condensers
15. **Zinc from Lead** Slags by Slag Fuming • Slag produced in the blast furnace have considerable amount of zinc in it • The slag is fed to a slag fuming furnace • The following reactions take place • The ZnO fumes are collected in bags and sent for Zn recovery to a zinc plant

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